

Gold-Free Faithfulness Metrics for NL-to-FOL Translation

Executive Summary

September 2026

Goal

This research investigated whether a candidate first-order logic (FOL) formula can be scored for faithfulness to its source natural-language sentence *without* a reference (gold) formula. The run explored three metric families—deterministic formula checks (FOL-Triage), cross-family solver consensus, and text-based alignment—over five iterations, benchmarked against LLM judges and structural baselines on two public datasets (FOLIO, MALLS) with solver-verified labels. The bar to beat was a disguised cheap LLM judge (Gemini 2.5 Flash Lite, AUROC 0.777).

What was tried

Iteration 1. Four experiments on the Logic-LM frozen screen ($n = 546$): FOL-Triage (deterministic lint, bag-of-words, and questionnaire layers), cross-system consensus via z3 equivalence among 9 LLM families, a full baselines suite (judges, round-trip NLI, structural metrics), and a summary comparison of discrimination vs. rename invariance.

Iteration 2. Five experiments on development set E ($n = 2,686$ rows, 292 sentences): PEER+TEXT fusion of consensus with text layers, re-evaluation under alternative label regimes, name-free consensus, and baselines. Also attempted: B2 decomposed z3 reader (DROPPED) and R_ADJ adjudicator (DROPPED).

Iteration 3. Confirmation on development set E with the head-on API bar test (T1), long composed sentences from R_COMP, PERTURB sensitivity/rename tests, and mechanism verification (hypothesis testing on error endorsement and complexity slope).

Iteration 4. Candidate-Signature Consensus (CSC): tested whether giving peers the candidate’s own symbols improves discrimination. Also: CSC on PERTURB (budget exhausted), E2 confirmation set construction (partial), R_COMP FREE name-free labels, and vocabulary analysis.

Iteration 5. Freeze-and-screen of five V0 variants, Gloss-Gated (GG) name-free checker, and three confirmation arms (all NOT_TESTABLE due to budget exhaustion).

Key results

The cross-family solver consensus metric (c_score_align) is the strongest gold-free signal discovered. It beats the cheap judge by a statistically significant margin on both the frozen screen and the development set. Adding text-accounting layers (PEER+TEXT) provides no significant gain over c_score_align alone once stratum-composition confounds are removed. The consensus advantage is

Result	Source	AUROC	95% CI	Notes
c_score_align (consensus)	Iter. 1, Exp. C	0.866	[0.821, 0.906]	Pooled, track L ($n = 546$)
c_score_align (stratified)	Iter. 3, Test T1	0.774	[0.739, 0.808]	Dev set E, R_AB PRIMARY
Δ vs judge (stratified)	Iter. 3, Test T1	+0.099	[+0.049, +0.146]	Paired, flash-lite bar
PEER+TEXT fusion	Iter. 2, Exp. 5	0.790	[0.75, 0.82]	R_AB pooled ($n = 2,686$)
Fusion – c_align (strat)	Iter. 2, Exp. 5	+0.011	[−0.012, 0.038]	Not significant
FOL-Triage fused	Iter. 1, Exp. A	0.759	[0.695, 0.825]	Below judge bar
Cheap judge (flash-lite)	Iter. 1, Exp. D	0.777	[0.720, 0.827]	The bar ($n = 524$)
SIGPROXY (cued, 3-fam)	Iter. 4, Exp. 10	0.930	[0.916, 0.943]	R_COMP, controlled vocab

largest on long, exception-bearing sentences: on R_A L20+EXC, c_score_align reaches 0.860 while the judge falls to 0.554 ($\Delta +0.306$ [+0.232, +0.376]).

What failed

Dead end	Iter.	Reason
L1 lint on LLM outputs	1	AUROC 0.508; does not transfer from human annotation errors
L2 role-aware check	1	AUROC 0.557; below simpler bag-of-words
Pilot structural metrics	1	AUROC 0.500–0.533 (null)
Round-trip reformalisation	1	AUROC 0.513; 76% ties
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476 (target ≥ 0.85)
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662 (target ≥ 0.80)
Name-free consensus (c_nf)	2–3	SWAP recall 0.512; sacrifices structural-error detection
CSC (Candidate-Signature)	4	Strat AUROC 0.614; anchoring $e = 0.459$ (gates G1, G2 fail)
GG (Gloss-Gated) checker	5	Strat AUROC 0.696; $d = 0.629$ (gate g3 FAIL)

The fundamental tradeoff is between discrimination and rename invariance. Consensus achieves the highest AUROC (0.866) but has a RENAME flip rate of 0.765 because z3 equivalence is sensitive to predicate names. FOL-Triage is nearly rename-invariant (flip rate 0.013) but sits below the judge bar. CSC attempted to fix the rename problem by sharing the candidate’s vocabulary with peers; the anchoring effect ($e = 0.459$) dominated the convergence benefit (d reduction -0.184), producing a net negative. GG attempted to bypass names entirely with a gloss-gated meaning matcher; it achieved low error endorsement ($e = 0.081$) but very high correct-row divergence ($d = 0.629$).

Best result

Cross-family solver consensus (c_score_align): **stratified AUROC 0.774** [0.739, 0.808] on development set E (R_AB PRIMARY, 354 rows, 144 sentences), confirmed in iteration 3, Test T1. It

beats the flash-lite judge by $\Delta +0.099$ [$+0.049$, $+0.146$] (paired, stratified). On long sentences with exceptions, the advantage grows to $\Delta +0.306$. The metric costs \$0.000525 per candidate ($30\times$ cheaper than the judge) and requires peer translations from ≥ 3 independent model families plus z3 pairwise equivalence checking. Its main limitation is non-invariance under predicate renaming (flip rate 0.765 under RENAME); all attempts to fix this (name-free consensus, CSC, GG) degraded discrimination.

Open questions

1. **Rename invariance.** No variant achieved both high AUROC and low rename sensitivity. Can a normalisation layer (canonical naming, signature-free equivalence) close the gap without anchoring?
2. **Confirmation on held-out data.** The E2 confirmation set was constructed (550 sentences) but labelling was incomplete due to budget exhaustion. All three iteration-5 confirmation arms are NOT_TESTABLE.
3. **Flash-lite bar on E.** The primary bar (flash-lite judge) is UNTESTABLE on the full development set; only 20 items were scored before API key exhaustion. The local Qwen3-8B fallback was used instead.
4. **Frontier judge interaction.** Under adjudicated labels, the frontier judge advantage over the cheap judge is $+0.166$ (significant). Whether consensus still beats the frontier judge on adjudicated labels is untested.
5. **Cost of diversity.** The peer pool requires ≥ 3 model families. Whether cheaper or fewer peers suffice without degrading AUROC is unexplored.
6. **Error typing.** Consensus detects COMPOUND and polarity errors well (recall 0.55, 0.53) but has near-zero recall on MEANING.RENAME and PEER.ENDORSED errors. Can error-type-specific signals be combined?